

DIGITAL FORENSICS LABORATORY · LAB 02

Common Windows Artifacts

Working paper — registry, LNK shortcuts, browser stores, and event logs

Name	Student ID (NIM)	Class / Group
------	------------------	---------------

Date	Instructor / TA	Session grade
------	-----------------	---------------

1. Lab Overview

This working paper accompanies Lab 02 of the digital forensics series. Windows itself is the primary witness in this lab: its Registry, shortcut files, browser stores, and event logs record who did what, when, and where. Complete the pre-lab before the session, check off each step as you perform it, answer every question with numbered evidence, and close with a reflection.

1.1 Learning objectives

- Explain what the Windows Registry is and name the five root keys (hives).
- Analyze NTUSER.DAT with Registry Explorer: settings, typed paths, autostart, USB mounts, program execution.
- Recognize persistence mechanisms in Run/RunOnce and explain why StartupApproved matters.
- Decode UserAssist ROT13 entries and explain what they prove.
- Parse LNK shortcuts with LECmd and list what a shortcut remembers.
- Query browser stores (places.sqlite) with SQL and decrypt saved logins with `firefox_decrypt`.
- Read event logs: filter by Event ID, hunt PowerShell 4104 script blocks, and rebuild an attack timeline.

1.2 Environment and files

Environment: a Windows VM or host (for your own machine you only READ - never modify), plus Kali Linux or any machine with Python for `firefox_decrypt` and `sqlite3`. Copy every evidence file to your working directory first - the live NTUSER.DAT is locked by Windows, and you never analyze originals in place.

Table 1.1 — Lab files

File	Purpose
files/NTUSER.DAT	Challenge A — registry hive of a compromised user (upstream)
files/Firefox.zip	Challenge A — Firefox profile of a suspect (upstream)
files/Microsoft-Windows-PowerShell%40Operational.evtx	Challenge A — PowerShell event log (upstream)
files/file.ps1	Challenge A — the script the attacker downloaded (upstream)
files/suspect_ntuser.reg	Challenge B (new) — exported hive of an insider suspect
files/places.sqlite	Challenge C (new) — 310 rows of browsing history with one suspicious evening

1.3 What to submit

- Section 1.4 pre-lab, answered before the session (worth 10 points).
- This working paper as PDF, named NIM_Name_Lab02.pdf.
- Your evidence folder NIM_Lab02/evidence/ containing every numbered exhibit (B-01, B-02, and so on).
- Answers that cite exact key paths, file names, Event IDs, or SQL - not vague descriptions.

1.4 Pre-Lab Check

Answer these before the session, using the lab README and Section 2 of this paper as references. One or two sentences per question are enough - precision beats length.

P1. What is the Windows Registry, and where does Windows store the per-user portion of it?

Answer:

Two sentences maximum.

P2. What does it mean when malware achieves persistence through the Run key? Describe in one sentence.

Answer:

Two sentences maximum.

P3. Name two artifacts that record which USB device was plugged into a machine.

Answer:

Two sentences maximum.

P4. Which file holds Firefox browsing history, and what tool opens it?

Answer:

Two sentences maximum.

P5. What happens on Event ID 4625, and why would one hundred of them in one minute matter?

Answer:

Two sentences maximum.

2. Theory Primer

2.1 The Registry and its hives

The Registry is a hierarchical database storing configuration for users, applications, and hardware. It is organized as a tree: keys (like folders) contain subkeys and values (name = data), and value types include REG_SZ (text), REG_DWORD (32-bit number), and REG_BINARY (raw bytes). Five root keys sit at the top.

Table 2.1 — The five root keys

Root key	Holds	Forensic interest
HKEY_CURRENT_USER	configuration of the logged-in user	most user traces; loaded from NTUSER.DAT
HKEY_USERS	all loaded user profiles	compare profiles, spot the odd one
HKEY_LOCAL_MACHINE	system-wide configuration	services, machine-wide autostart
HKEY_CLASSES_ROOT	file-type associations	how a payload executes on open
HKEY_CURRENT_CONFIG	current hardware profile	display and docking details

2.2 NTUSER.DAT and the forensic hot spots

Each user's hive is stored as C:\Users\%USERNAME%\NTUSER.DAT. It is locked while the user is logged on, so analysis is done on a copy using Registry Explorer. The table below lists the locations that repay a visit on every case.

Table 2.2 — Hot spots under HKEY_CURRENT_USER

Location	Records	Watch out for
Software\Microsoft\Windows\CurrentVersion\Run	programs starting at every logon (persistence)	names imitating Microsoft products
...\RunOnce + the HKLM twins	one-shot and machine-wide autostart	always check all four Run scopes

Location	Records	Watch out for
...\Explorer\StartupApproved\Run	which Run entries are enabled/disabled	a freshly toggled entry = recent change
...\Explorer\TypedPaths	paths typed into Explorer, url1 = most recent	reveals where the user navigated
...\Explorer\UserAssist	GUI program executions with run counts	value names are ROT13-encoded
...\Explorer\MountPoints2	mounted volumes including USB devices	USB serial number survives reformatting
...\Explorer\RecentDocs	recently used documents (MRU, binary)	use a parser to read

2.3 LNK shortcuts and LECmd

A .lnk shortcut is created automatically when a file is first opened, and it remembers more than most users imagine: the target's full path and timestamps, the working directory, the hostname, and even the MAC address of the machine that created it. LECmd (Eric Zimmerman) turns a .lnk into a readable report. Look for shortcuts in the Recent folder, Desktop, and Start Menu.

2.4 Browser stores

Firefox keeps everything in one profile folder under AppData\Roaming\Mozilla\Firefox\Profiles. Chrome keeps its data under AppData\Local\Google\Chrome\User Data\Default, with the encryption key in the Local State file. Both store history in SQLite - which means the full power of SQL applies.

Table 2.3 — Firefox profile files that matter

File	Holds	Read with
places.sqlite	history, bookmarks, downloads	DB Browser for SQLite / sqlite3
key4.db + logins.json	master key + encrypted saved logins	firefox_decrypt
cookies.sqlite	session cookies	DB Browser for SQLite
formhistory.sqlite	search box and form entries	DB Browser for SQLite

Firefox timestamps are PTime: microseconds since 1601-01-01. Convert with `datetime(visit_date/1000000, 'unixepoch')` in SQL, or subtract 11644473600 seconds after dividing by one million.

2.5 Event logs

Windows records system, security, and application events in .evtx files under C:\Windows\System32\winevt\Logs. Every event carries an Event ID - learning the important IDs turns gigabytes of noise into a searchable narrative. Explore with Event Viewer; query with Get-WinEvent or wevtutil.

Table 2.4 — Event IDs worth memorizing

Event ID	Meaning
4624 / 4625	logon SUCCESS / FAILURE (brute force pattern)
4634 / 4647	logoff
4648	logon with explicit credentials (lateral movement hint)
4688	a new process was created
4720 / 4726	user account created / deleted
1102	the audit log was CLEARED - classic anti-forensics
1116 / 1117	Defender detected malware / took action
4104	PowerShell script block - the script text itself is logged

The PowerShell Operational channel with Event ID 4104 (script block logging) is one of the highest-value logs in incident response: when enabled, it records the actual script text the attacker executed - downloads, deletions, encoded payloads.

3. Guided Exercises

Work through the five drills in order. Every output is an exhibit: capture it, number it (B-01, B-02, and so on), and log it in Appendix B.

3.A Registry recon (NTUSER.DAT)

- 1. Copy NTUSER.DAT to your working directory; compute and record its MD5 and SHA1.
- 2. Open the copy in Registry Explorer and navigate to Control Panel\Mouse. What is DoubleClickSpeed?
- 3. Navigate to Software\Microsoft\Windows\CurrentVersion\Explorer\TypedPaths. What is the most recent typed path and how do you know?
- 4. Navigate to Software\Microsoft\Windows\CurrentVersion\Run. List every value, mark the legitimate ones, and identify the persistence mechanism.
- 5. Record each finding as a numbered exhibit showing the key path and the value.

3.B Hot-spot sweep

- 1. Check StartupApproved\Run under the same hive: are all Run entries enabled?
- 2. Look at Explorer\UserAssist - do the value names look readable? Explain why not, and what they represent.
- 3. Look at Explorer\MountPoints2 - list the mounted volumes; identify any USB storage device by serial number.
- 4. Record your observations, citing full key paths.

3.C Firefox analysis

- 1. Unzip Firefox.zip to your working directory (never modify the archive).
- 2. Open the profile's places.sqlite in DB Browser for SQLite; list the tables.
- 3. Rank visited domains by URL count (visit_count alone can mislead - explain why grouping by domain is safer).
- 4. Rebuild the last five visits in time order, converting PRTime to a readable date.
- 5. Optional (needs Python): run firefox_decrypt against the profile and record the saved login.

3.D The PowerShell log

- 1. Open the PowerShell Operational evtx in Event Viewer; filter for Event ID 4104.
- 2. List the script blocks in chronological order and describe the attacker's sequence of actions.
- 3. Which command downloaded a file, and what was the file named?
- 4. Open file.ps1 (from files/) and explain what it does - decode anything encoded.

3.E Your own machine (read-only!)

- 1. Locate your own NTUSER.DAT path and your Recent folder (do NOT modify or delete anything).
- 2. Read-only query: reg query "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" - list what starts at YOUR logon.
- 3. Count the .lnk files in your Recent folder; open one with a text viewer and note that most bytes are binary.
- 4. Reflect: which artifact on your own machine surprised you the most?

4. Challenges

Challenge A is the upstream case - solved together in the Deep Dive session. Challenges B and C are yours: work on copies, hash before and after, cite key paths and SQL. Flags use ctf{xxxx} (Challenge A keeps the upstream flag format). Hints are staged - stop after each one and retry.

4.1 Challenge A — the upstream trio (solved in the Deep Dive)

The three evidence sets from Table 1.1: NTUSER.DAT, the Firefox profile, and the PowerShell log. The Deep Dive deck walks through the full solution - use your own exhibits, not the deck's screenshots.

4.2 Challenge B — the exported hive (files/suspect_ntuser.reg)

- 1. An insider suspect's hive was exported to suspect_ntuser.reg (plain text). Sweep it with your Lab 01 skills (grep/Select-String) and a text editor.

- **2.** List every value under the Run key. One of them is persistence wearing a legitimate-looking name - which one, and what two details give it away?
- **3.** Decode the encoded command. What would run at every logon, and what does it reveal?
- **4.** The UserAssist entries are ROT13 - decode them and name the two interpreters the suspect executed.
- **5.** What was the most recent path the suspect typed, and which USB device (vendor, model, serial) was mounted?

Hints: (1) The suspicious Run value hides among three legitimate-looking ones - read the data column, not just the names. (2) -enc means encodedcommand; the blob is base64 of UTF-16LE text. (3) ROT13 is a rotation, not encryption - any online or one-line decoder works.

Flag B: ctf{_____}

4.3 Challenge C — the history hunt (files/places.sqlite)

- **1.** Copy places.sqlite to your working directory and open it with DB Browser for SQLite (or sqlite3).
- **2.** Rank the visited domains by URL count. Which domain dominates the history?
- **3.** Convert PRTIME and rebuild the evening of 20 September 2026 (21:00-23:00) as an ordered timeline.
- **4.** Tell the story: what did the user search for, read, copy, and download that evening?
- **5.** One search in that evening contains a URL-encoded secret. Find it and decode it.

Hints: (1) Domains live between // and the next / - GROUP BY a computed host works, or export and use awk. (2) datetime(visit_date/1000000,'unixepoch') makes PRTIME human. (3) LIKE '%ctf%' finds the row; %7B and %7D are the encoded braces you last met in Lab 01.

Flag C: ctf{_____}

5. Analysis Questions

Questions 1-3 come from the original lab; questions 4-12 extend the case. Cite exact key paths, values, Event IDs, or SQL in every answer.

1. Given the registry file (NTUSER.DAT) of a compromised system:

- a.** What is the mouse double-click speed?
- b.** What is the most recent typed path recorded in the registry?
- c.** Which value did malware.exe add to establish persistence? Give the key path, value name, and data.

Answer:

Write your complete answer, including the exact key path, file, or command used.

Supporting evidence (screenshot / output): exhibit numbers B-____ to B-____

2. Given the Firefox profile of the suspect:

- a.** Which username and password are stored in the saved logins?
- b.** Which website was visited most frequently? Explain how you determined it.

Answer:

Write your complete answer, including the exact key path, file, or command used.

Supporting evidence (screenshot / output): exhibit numbers B-____ to B-____

3. Given the PowerShell event logs of a compromised system:

- a.** Which command did the attacker use to download a file onto the system?
- b.** Analyze the downloaded file - what is its purpose, down to its final output?

Answer:

Write your complete answer, including the exact key path, file, or command used.

Supporting evidence (screenshot / output): exhibit numbers B-____ to B-____

4. Why is the Run key such a popular persistence mechanism, and which OTHER autostart location in Table 2.2 would you check next? Justify your choice.

Answer:

Write your complete answer, including the exact key path, file, or command used.

Supporting evidence (screenshot / output): exhibit numbers B-____ to B-____

5. UserAssist stores program names in ROT13. Explain why this is obfuscation rather than protection, and what the entries still prove.

Answer:

Write your complete answer, including the exact key path, file, or command used.

Supporting evidence (screenshot / output): exhibit numbers B-____ to B-____

6. A colleague reports `visit_count = 1` for every row of a `places.sqlite` and concludes 'no favorite sites'. What is the flaw in that reasoning, and which query gives the real answer?

Answer:

Write your complete answer, including the exact key path, file, or command used.

Supporting evidence (screenshot / output): exhibit numbers B-____ to B-____

7. Convert the PRTIME value 13434416546000000 to a readable UTC date-time. Show your formula.

Answer:

Write your complete answer, including the exact key path, file, or command used.

Supporting evidence (screenshot / output): exhibit numbers B-____ to B-____

8. An attacker deleted the registry value pointing to their tool. Name two ways the value might still be recovered from the system.

Answer:

Write your complete answer, including the exact key path, file, or command used.

Supporting evidence (screenshot / output): exhibit numbers B-____ to B-____

9. During a review you find Event ID 1102 followed by 4624 logons of a new account. What does this sequence suggest, and what would you check next?

Answer:

Write your complete answer, including the exact key path, file, or command used.

Supporting evidence (screenshot / output): exhibit numbers B-____ to B-____

10. Why must analysis be performed on copies of NTUSER.DAT and the evtx files rather than the originals? Relate your answer to chain of custody.

Answer:

Write your complete answer, including the exact key path, file, or command used.

Supporting evidence (screenshot / output): exhibit numbers B-____ to B-____

11. Which single artifact from this lab would you present first to a manager with no technical background, and why?

Answer:

Write your complete answer, including the exact key path, file, or command used.

Supporting evidence (screenshot / output): exhibit numbers B-____ to B-____

6. Reflection and Self-Assessment

Summarize your main findings, the skills that clicked, and the obstacles you hit - with how you solved them.

Reflection:

Findings, new skills, obstacles, and how you overcame them.

Supporting evidence (screenshot / output): exhibit numbers B-____ to B-____

Table 6.1 — Self-assessment checklist

I can	Yes	Not yet
Navigate the Registry and explain the five hives		
Find persistence in Run/RunOnce and check StartupApproved		
Decode UserAssist ROT13 and explain what it proves		
Parse a .lnk with LECmd and list what it reveals		
Query places.sqlite with SQL and convert PRTIME		
Hunt PowerShell 4104 script blocks and rebuild a timeline		

7. Appendices

Appendix A — Tool cheat sheet

Task	Command / tool
Query a live registry key	<code>reg query "HKCU\...\Run" (add /s to recurse)</code>
Export a key to text	<code>reg export HKCU\...\Run backup.reg</code>
Open an offline hive	Registry Explorer (Zimmerman)
Parse a shortcut	<code>LECmd.exe -f file.lnk</code>
Open a SQLite store	DB Browser for SQLite, or <code>sqlite3 places.sqlite</code>
Human-readable PRTIME	<code>SELECT datetime(visit_date/1000000,'unixepoch');</code>
Decrypt Firefox logins	<code>python firefox_decrypt.py <profile-folder></code>
Last events of a channel	<code>Get-WinEvent -LogName '...' -MaxEvents 20</code>
Filter by Event ID	<code>Get-WinEvent -FilterHashtable @{LogName='...'; Id=4104}</code>
Classic evtq query	<code>wevtutil qe Security /q:"*[System[(EventID=4625)]]" /c:10</code>

Appendix B — Evidence log

One row per exhibit. Keep numbering continuous across the whole session.

What good evidence looks like — model exhibit:

```
[B-05] 2026-09-30 09:41 Registry Explorer
HKCU\Software\Microsoft\Windows\CurrentVersion\Run
    Malware REG_SZ C:\Users\w\Desktop\malware.exe
-> captured as evidence/B-05_run_malware.png
```

- The key path AND the value (name, type, data) are fully visible.
- The tool and moment of capture are recorded; the filename says what it shows.
- One exhibit supports one finding.
- Bad evidence: a screenshot of a tree with no visible path bar, or an answer with no exhibit number.

No.	Time	Activity / Command	Source file / Tool	Notes
B-01				
B-02				
B-03				
B-04				
B-05				
B-06				
B-07				
B-08				
B-09				
B-10				

Appendix C — Hash verification form

Fill one row per evidence file: hash when received (start) and after analysis (end). Identical values prove faithful handling.

No	Evidence file	Algorithm	Hash (start)	Hash (end)	Match?
1					Yes / No
2					Yes / No
3					Yes / No

No	Evidence file	Algorithm	Hash (start)	Hash (end)	Match?
4					Yes / No
5					Yes / No
6					Yes / No

Appendix D — Glossary

Term	Meaning in this lab
registry / hive	Windows' hierarchical configuration database; a hive is one on-disk file of it
key / value	a node in the registry tree; a named data entry inside a key (REG_SZ, REG_DWORD, REG_BINARY)
NTUSER.DAT	the per-user hive: settings, typed paths, autostart, USB mounts, program use
MRU	Most Recently Used ordering - url1 in TypedPaths is the newest entry
persistence	a mechanism that keeps malware running after reboots, e.g. a Run-key value
ROT13	a letter rotation (A to N); UserAssist program names are stored this way
SID	Security Identifier - Windows' internal account identity
LNK	a shortcut file; remembers target path, timestamps, hostname, MAC address
PRTIME	Firefox timestamp format: microseconds since 1601-01-01
sqlite	an embedded SQL database engine; browsers store history in its files
key4.db / logins.json	Firefox master-key store and encrypted saved logins
event log (.evtx)	Windows' binary event records under winevt\Logs
Event ID	the numeric code identifying what an event means (4624, 4104, and so on)
channel	a named event stream, e.g. Windows PowerShell/Operational
script block logging	PowerShell feature (Event 4104) recording executed script text
anti-forensics	attacker actions that destroy evidence, e.g. clearing logs (Event 1102)
lateral movement	an attacker spreading from one machine to another (hint: Event 4648)
working copy	the copy you analyze; originals stay sealed and hashed

Honesty statement: by submitting this working paper I declare that all answers and evidence are my own work, that original evidence files were never modified, and that I followed the lab's ethics rules.

Student signature: _____ Date: _____

Instructor / TA signature: _____ Grade: _____